LOGICAL DESIGN PROJECT

Team Contributions, Challenges & Modeling Decisions

1. Team Members and Roles

Aouatif Guennoun

- Group leader, responsible for managing the GitHub repository.
- Responsible for writing the report and identifying the issues encountered during the project.

Valdez Kemekong

- Contributed to finding solutions whenever the group encountered difficulties by providing real-world examples.
- Contributed to the development of the SQL code.

Ranya El Bahali

- Discussed the different possible solutions for each task with the team members.
- Responsible for writing the report and identifying the issues encountered during the project.

Hayat Ennani

- Shared suggestions with the team regarding each task and the most efficient ways to approach them.
- Participated in preparing the project report and provided its final version.
- Discussed the different possible solutions for each task with the team members.

2. Key Challenges and How They Were Addressed

Aggregation in the Logical Design

We encountered difficulty in identifying the primary key of the “CONSUME” relationship. In particular, we were unsure whether the combination (reservation_id, consumable_id, laboratory_id) should be used as the final primary key.

ISA Representation

We initially had difficulty distinguishing situations in which a separate table should be created for a child entity from situations in which the child entity should be merged with its parent table.

3. Modeling Choices and Justifications

Aggregation Problem

We were concerned that a user might modify the table in a way that would leave Laboratory_Id and Consumables_Id unrelated through the “STOCKS” relationship. After discussing the issue, we determined that the SQL constraints automatically prevent such an inconsistent state by returning an error.

ISA Problem

We discussed the possible solutions and concluded that merging the child entity into the parent relation depends on the context and requirements of the database.